Class 7
CHEMISTRY

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1. ATOMIC STRUCTURE KEY & SOLUTIONS

WORKSHEET - 1

Level – I

1) A	2) B	3) D	4) A	5) C	6) C
7) C	8) C	9) A	10) B		

Level – II

1) D	2) D	3) A	4) A	5) D	6) A
7) B	8) B	9) D	10) B	11) A	12) B
13) D	14) B	15) D	16) A		

Level – III

17) A,B,C,D	18) B,D	19) B	20) C	21) B	22) A
23) a-q, b-r,c-p,d-s		24) A,C	25) B	26) A	27) C
28) A,B,C,D					

WORKSHEET – 2

Level – I

1) A	2) A	3) C	4) B	5) C	6) A
7) A	8) C	9) C	10) B	11) D	12) C
13) B					

Level – II

1) A	2) B	3) C	4) A	5) A	6) C
7) B	8) C	9) C	10) A	11) B	12) B
13) D					

Level – III

14) A,B,C 15) A,C 16) A 17) D 18) C 19) B

20) a-q; b-s; c-p; d-r 21) 0 22) A,B,C,D

23) a-r; b-p; c-q; d-s 24) 4 25) A 26) A 27) A

WORKSHEET – 3

Level – I

1) D 2) A 3) A 4) A 5) B 6) A

7) B 8) A 9) D 10) B 11) A 12) A

13) D

Level – II

1) A 2) D 3) B 4) C 5) D 6) C

7) D 8) B 9) D 10) C 11) B 12) D

13) D 14) C 15) D

Level – III

16) A,B 17) A, C 18) D 19) C 20) C 21) B

22) C 23) A 24) a -3, b - 4 , c -2, d - 1 25) 0 26) A, B, C

WORKSHEET – 4

Level – I

1) A 2) B 3) A 4) C 5) A 6) B

7) C 8) B

Level – II

1) C 2) C 3) C 4) A,B,C,D 5) A 6) B

7) B 8) D 9) C 10) B 11) A 12) A

13) B

Level – III

14) A,B,C,D 15) A 16) B 17) B 18) D

19) a -q, b- r, c- p, d- s 20) +1 or + 1.602 ×k-19c or +4.8 ×10-10 esu

21) A,B 22) A 23) B 24) A 25) C 26) B

27) A

WORKSHEET – 5

Level – I

1) A 2) A 3) C 4) C 5) A 6) C

7) A 8) A 9) C 10) A

Level – II

1) C 2) D 3) A 4) C 5) C 6) A

7) C 8) B 9) C 10) A 11) A 12) A

13) C 14) D 15) D 16) D

Level – III

17) A,B,C 18) A 19) C 20) D 21) A

22) a -4, b -1, c -2, d – 3 23) 1 24) A,B,C,D 25) A 26) A

27) B

WORKSHEET – 6

Level – I

1) B 2) D 3) D 4) C 5) D 6) A

7) C 8) C 9) C 10) B 11) A 12) B

13) B 14) A 15) C 16) C 17) C

Level – II

1) B 2)A 3) B 4)C 5) D 6) C

7) B 8) C 9) C 10) D 11) C 12) C

13) B 14) A 15) B 16) C 17) A 18) A

19) A 20) C

Level – III

21) A,B,D 22) D 23) A 24) C 25) a-s, b-r, c-t, d-q

26) A,B,C 27) A, B,C

WORKSHEET – 7

Level – I

1) A 2) A 3) B 4) B 5) C 6) B

7) D 8) A

Level – II

1) A 2) A 3) C 4) D 5) D 6) B

7) D 8) B 9) C 10) B 11) $10A^0$ 12) $C = \nu\lambda$

2. Wave length $(\lambda) = 2\ A^0$

Same electromagnet Radiation wave length in centimetre $= i\lambda, 1\ A^0 =$

10^{-8} cm $2\ A^0 = 2 \times 10^{-8}$ cm

5. Frequency (v) $= 6 \times 10^{15}\ s^{-1}$

Wave number $(\bar{v}) = \frac{v}{c}$

$$2 \times 10^{15} \times 10^{-10} = \frac{6 \times 10^{15}\ s^{-1}}{3 \times 10^{10}\ cm \cdot s^{-1}}$$

$(\bar{v}) = 2 \times 10^{5}\ cm^{-1}$

6. Wave length $O(\lambda) = 400$ nm

$$v = \frac{C}{\lambda}$$

$$v = \frac{3 \times 10^{8}\ cm \cdot s^{-1}}{4 \times 10^{-7}\ cm}$$

$\Rightarrow 0.75 \times 10^{15}\ s^{-1}$

$\Rightarrow 7.5 \times 10^{14}\ s^{-1}$

7. $\bar{v} = 1 \times 10^{-5}\ \text{cm}^{-1}$

$v = C\bar{v}$

$= 3 \times 10^{10}\ \text{cm} \cdot \text{s}^{-1} \times 1 \times 10^{-5}\ \text{cm}^{-1}$

$= 3 \times 10^{5}\text{sec}^{-1}$

8. $v = 4.69 \times 10^{14}\text{sec}^{-1}$

$\lambda = \frac{C}{v}$

$= \frac{3 \times 10^{8}\ \text{m} \cdot \text{sec}^{-1}}{4.69 \times 10^{14}\text{sec}^{-1}}$

$= 6.39 \times 10^{-7}\ \text{m}$

$1\ \text{A}^\circ = 10^{-10}\ \text{m}$

$= 6.39 \times 10^{3}\ \text{A}^\circ$

$= 6390\ \text{A}^\circ$

9. $\lambda = 400\ \text{nm}$

$1\ \text{nm} = 10\ \text{A}^\circ$

$400 \times 10\ \text{A}^\circ = 4000\ \text{A}^\circ$

10. $T = 2.0 \times 10^{-10}\ \text{s}$

$T = \frac{1}{v}$

$vT = 1$

$v = \frac{1}{T} = \frac{1}{2.0 \times 10^{-10}\ \text{s}}$

$= 0.5 \times 10^{10}\text{sec}^{-1}$

(or)

$= 5 \times 10^{9}\text{sec}^{-1}$

Level – III

13) A 14) C 15) C 16) A,D 17) C,D 18) D

19) A 20) B 21) D,C,B,A 22) A,C 23) A,B,C,D 24) A,B,C,D

25) A,B,C,D 26) a-q,b-r.c-s,d-p

WORKSHEET – 8

Level – I

1) B 2) B 3) B 4) C 5) C 6) A

7) B 8) B 9) B

Level – II

1) A 2) A 3) A 4) D 5) B 6) D

7) C 8) B 9) D 10) A 11) D 12) C

13) D 14) D 15) C 16) A

Level – III

17) B 18) B,D 19) A,C 20) B 21) D 22) C

23) a-s, b-p, c-q, d-r 24) A,B 25) A,C,D 26) a-p, b-q,c-s,d-r

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WORKSHEET – 9

Level – I

1) B 2) A 3) B 4) A 5) B 6) B

7) C 8) A

Level – II

1) B 2)C 3) B 4) D 5) C 6) B

7) B 8) B 9) C 10) D 11) B 12) A

13) D 14) D 15) C 16) A 17) B 18) 12

Level – III

19) A,B 20) A,B,C,D 21) A 22) B 23) C

24) a-4,b-2,c-3,d-1 25) B,D 26) A,C 27) A,C

28) a-p,q,r,t; b-p,q,t; c- p,q,t; d-r

2. PERIODIC PROPERTIES

WORKSHEET-1

Level - I

1) C 2) B 3) A 4) A 5) B 6) C 7) D
8) A 9) B 10) B

Level – II

1) A 2) B 3) D 4) A 5) C 6) D 7) B
8) B 9) D 10) B 11) D 12) D 13) C 14) B
15) A 16) A 17) B 18) A

Level – III

19) A,B,C 20) D 21) D 22) C 23) a-p ; b-q ; c-r,s ;d-t
24) C,D 25) a-q; b-r ; c-r ;d-s 26) A,B,C 27) a-p ; b-r ; c-s ;d-q

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WORKSHEET-2

Level – I

1) B 2) B 3) A 4) A 5) A 6) B 7) C
8) D 9) C 10) A 11) A

Level – II

1) B 2) C 3) A 4) B 5) B 6) B 7) C
8) C 9) C 10) A 11) C 12) C 13) C 14) C
15) D 16) A

Level – III

17) A,B,C 18) B 19) A 20) A 21) a-t ; b-r ; c-q; d-p 22) C
23) D 24) C 25) C 26) a-s ; b-t ; c-q; d-p e-r; 27) D
28) a-r ; b-s ; c-q; d-p

WORKSHEET-3

Level – I

1) A 2) D 3) A 4) A 5) B 6) D 7) D

Level – II

1) B 2) A 3) B 4) C 5) 3 6) C 7) C
8) A,B,D 9) 0 10) B 11) C 12) C 13) 14) 6
15) C 16) A 17) B 18) A,B

Level – III

19) D 20) B 21) A 22) a-s ; b-q ; c-r ;d-p 23) C 24) B
25) A 26) B 27) A 28) a-s ; b-r ; c-q ;d-p

WORKSHEET-4

Level – I

1) C 2) A 3) C 4) B 5) C 6) A

Level – II

1) C	2) B	3) B	4) A	5) B	6) D	7) B
8) C	9) C	10) A	11) C	12) B	13) D	14) B
15) A	16) C	17) D	18) B			

Level – III

19) A,C,D	20) C	21) A	22) D	23) a-q,p; b-s; c-p; d-r		
24) A	25) B,C	26) D	27) a-r; b-p;c-q; d-s		28) A	29) B
30) a-q ; b-p,r,s,t; c-p ;d-q						

WORKSHEET-5

Level – I

1) D	2) D	3) B	4) A	5) B	6) D	7) D
8) D	9) C	10) A				

Level – II

1) D	2) A	3) A	4) A	5) B	6) B	7) C
8) A	9) D	10) A	11) D	12) B	13) B	14) C
15) A	16) D	17) B	18) A			

Level – III

19) A,B,C,D	20) B	21) A	22) C	23) a-t ; b-s ; c-p,q ;d-r	24) A,B
25) A,B,C,D	26) A,C	27) C,D	28) B	29) B	30) B

WORKSHEET-6

Level – I

1) A	2) A	3) B	4) A	5) A	6) A	7) A
8) B	9) A	10) D				

Level – II

1) B	2) A	3) A	4) B	5) D	6) C	7) D
8) B	9) A	10) A				

Level – III

11) A,B,C	12) D	13) A	14) A	15) a-p; b-q; c-p	
16) D	17) D	18) D	19) A	20) a-r; b-p; c-q; d-s	

WORKSHEET-7

Level – I

1) C	2) B	3) A	4) B	5) D	6) B	7) C
8) C	9) C	10) C				

Level – II

1) D	2) A	3) D	4) A	5) C	6) B	7) A
8) C	9) A	10) D	11) B	12) C	13) C	14) C
15) C	16) D	17) D	18) A	19) C	20) C	

Level – III

21) A,B,C 22) A,B,C,D 23) B 24) A 25) C 26) a-q; b-t; c-p; d-r
27) +2 28)+3 29) a-q,b-s,c-p,d-r 30) a-s; b-p; c-q; d-r
31) a- p,q,r; b-p,q,r; c-s,q; d-q

WORKSHEET-8

Level – I

1) C 2) B 3) A 4) B 5) A 6) C 7) B
8) D 9) B

Level – II

1) A 2) C 3) C 4) B 5) D 6) C 7) A
8) D 9) C 10) C 11) D 12) D 13) B 14) A

Level – III

15) A,B 16) A,B, C 17) A, C 18) A,D 19) D 20) A 21) B
22) a-r ; b-p ; c-q ;d-s 23) 1 24) a-s; b-r; c-p; d-q 25) a-s; b-r; c-q; d-p

3. MOLE CONCEPT

WORK SHEET-1

Level – I

1) B	2) D	3) B	4) B	5) D	6) B	7) A
8) C	9) B	10) B				

Level – II

1) A	2) C	3) C	4) A	5) D	6) A	7) A
8) A	9) D	10) B	11) B	12) B	13) B	14) C

6. $_{17}Cl^{37}$ mass no. = 37
   no. of protons = 17
   ∴ no. of electrons = no. of protons = 17.
7. $Mg(OH)_2$
   $1(Mg) + 2(O) + 2(H)$
   $24 + 2(16) + 2(1)$
   $24 + 32 + 2 = 58$
8. 1 mole of $CO_2 = 6.023 \times 10^{23}$ molecules
10. Molecular weight of homo atomic molecule is
    Atomic weight × atomicity
    Sulphur = $32 \times 8 = 256$ g
14. $200 \times \frac{90}{100} + 199 \times \frac{8}{100} + 302 \times \frac{2}{100} = 202$

Level – III

15) A, B	16) C	17) B	18) A	19) C	20) a-s; b-r; c-q; d-p
21) A, B, C	22) A, B, C	23) A	24) C	25) C	26) a-r; b-s ; c-q,s; d-p
27) A,B	28) a-q; b-p; c-r; d-s				

15. H_2SO_4
    $2(H) + S + 4(O) = 2(1) + 32 + 4(16)$
    $= 2 + 32 + 64; = 98$
    $H_3PO_4; 3(H) + 1(P) + 4(O) = 3(1) + 1(31) + 4(16)$
    $= 3 + 31 + 64; = 98$
19. 1 gram molecule of $CH_4 = 16$ g
    2 gram molecule of $CH_4 = 32$ g
22. $H_2 = 2(1) = 2$ g

$O_2 = 2(16) = 32$ g; $N_2 = 2(14) = 28$ g

27. 44 g of $CO_2 \rightarrow N_A$

4.4 g of $CO_2 \rightarrow x$

$x = \frac{4.4 \times N_A}{44} = 0.1N_A$

$28g$ of $N_2 \rightarrow N_A$

$2.8g$ of $N_2 \rightarrow x$

$x = \frac{2.8 \times N_A}{28} = 0.1N_A$

WORK SHEET-2

Level – I

1) D	2) A	3) B	4) A	5) C	6) D	7) C
8) C	9) A	10) D				

5. $Na_2CO_3 = 2(Na) + 1(C) + 3(O)$

$= 2(23) + 1(12) + 3(16)$

$= 46 + 12 + 48$

$= 106$ g

8. 56 g of iron → 1 mole

xg of iron → 7.4 mole

$x = 56 \times 7.4$

$= 414.4$ g

Level – II

1) A	2) D	3) D	4) C	5) D	6) D	7) A
8) C	9) C	10) B	11) B	12) A	13) D	14) A

1. 1 mole of iron → 6.023×10^{23} atoms

7.4 moles of iron → x

$x = 7.4 \times 6.023 \times 10^{23}$ atoms

$= 44.5 \times 10^{23}$

2. 56 g of Iron = 6.023×10^{23} atoms

1 g of Iron → x

$x = \frac{6.023 \times 10^{23}}{56} = 0.1075535 \times 10^{23}$

$= 1.07 \times 10^{22}$

9. $CH_4 = 1(C) + 4(H) = 1(12) + 4(1) = 12 + 4 = 16$ g

$NH_3 = 1(N) + 3(H) = 1(14) + 3(1) = 14 + 3 = 17$ g

$H_2O = 2(H) + 1(O) = 2(1) + 1(16) = 2 + 16 = 18$ g

10. $KClO_3 = 1(K) + 1(Cl) + 3(O)$

$= 39 + 35.5 + 3(16)$

$= 74.5 + 48 = 122.5$ g

1 mole of $KClO_3 = 122.5$ g

2 moles of $KClO_3 = 245$ g

12. $C_{60}H_{122}$

$60(12) + 122(1) = 842$ g

$6.023 \times 10^{23} \rightarrow 842$ g

1 molecule $\rightarrow$ xg

$x = \frac{842}{6.023 \times 10^{23}} = 1.4 \times 10^{-21}$ g

14. Molecular formula of Glucose : $C_6H_{12}O_6$

weight of Glucose : 180 g

6.023×10^{23} molecules of Glucose $\rightarrow$ 180 g of mass

1 molecules of Glucose $\rightarrow$ Xg

$X \times 6.023 \times 10^{23} = 180 \times 1$

$X = \frac{180}{6.023 \times 10^{23}}$

$= 29.8 \times 10^{-23}$

Level – III

15) B, D 16) A 17) A 18) C 19) a-s; b-r; c-q; d-p 20) B

21) B 22) B 23) C 24) A, B, C, D 25) a-q; b-p; c-r; d-s

15. 32 g of Oxygen $-N_A$ molecules

16 g of Oxygen $-N_A/2$

44 g of $CO_2 \rightarrow N_A$

22 g of $CO_2 \rightarrow N_A/2$

28 g of $CO \rightarrow N_A$

14 g of $CO \rightarrow N_A/2$

18. 1 mole of $CO_2 \rightarrow 44$ g

3 moles $CO_2 \rightarrow 3 \times 44 = 132$ g

20. $FeSO_4$

Fe: 56

S: 32

4(0): 64

152 g

21. 44 g of $CO_2 \rightarrow 22.4$ lit

22 g of $CO_2 \rightarrow x$

$x = \frac{22 \times 22.4}{44} = 11.2$lit

WORK SHEET-3

Level – I

1) A	2) B	3) B	4) D	5) A	6) B	7) B
8) D	9) C	10) B				

1. NO_2

$1(N) + 2(O) = 14 + 2(16) = 46$ g

2. 1 electron charge $= 1.602 \times 10^{-19}$C

1 mole of electron charge $= 6.023 \times 10^{23} \times 1.602 \times 10^{-19} = 96{,}500$C

2 mole of electron charge $= 2 \times 96500 = 193000$C

3. 2 gm molecule weight 64 g is Oxygen; $2(32) = 64$ g

4. 1gm molecule weight 64 g is SO_2

5. 8 g of $O_2 \rightarrow x$

32 g of $O_2 \rightarrow N_A$

$x = \frac{8 \times N_A}{32} = \frac{N_A}{4}$

7 g of $N_2 \rightarrow x$

28 g of $N_2 \rightarrow N_A$

$x = \frac{7 \times N_A}{28} = \frac{N_A}{4}$

9. 22.4 L of $O_2 \rightarrow 1$ mole

56 L°$O_2 \rightarrow x$

$x = \frac{56}{22.4} = 2.5$ moles

10. 1 mole $\rightarrow 22.4\ L_{of}\ CO_2$

1.5 mole $\rightarrow x$

$x = 22.4 \times 1.5 = 33.6L$

Level – II

1) A	2) B	3) A	4) B	5) C	6) B	7) D
8) B	9) B	10) D	11) A	12) C	13) A	14) A

1. 6.023×10^{23} atoms of Iron → 1 Mole

   4×10^{31} atoms of Iron → x

   $x = \frac{4 \times 10^{31}}{6.023 \times 10^{23}}$

   $= 6.6 \times 10^{7}$

2. 6.023×10^{23} molecules of Chlorine → 72 g

   6.4×10^{31} molecules of Chlorine → x g

   $x = \frac{72 \times 6.4 \times 10^{31}}{6.023 \times 10^{23}}$

   $= 76.508 \times 10^{8}$ g

3. 1 Mole of $Ca(OH)_2$ → 74 g

   4 Moles of $Ca(OH)_2 \rightarrow 4 \times 74 = 296$ g

4. $KClO_3: 39 + 35.5 + 48 = 122.5$ g

5. $2O_3: 2 \times 48 = 96$ g

6. 1 Mole of $CO_2 = 44$ g

   3.1 Moles of $CO_2 = 44 \times 3.1$

   $= 136.4$ g

8. 6.023×10^{23} atoms → 22.4 L of volume

   9.03×10^{23} atoms → xL of volume; $x = \frac{22.4 \times 9.03 \times 10^{23}}{6.023 \times 10^{23}} = 33.6$ L

9. 2 g of Hydrogen → 6.023×10^{23} molecules; 20 g of Hydrogen → x

   $x = \frac{20 \times 6.023 \times 10^{23}}{2} = 6.023 \times 10^{24}$

10. 64 g of SO_2 → 22.4 L

    320 g of $SO_2 \rightarrow x$ L

    $x = \frac{320 \times 22.4}{64} = 112L$

11. 6.023×10^{23} molecules → 100 g of $CaCO_3$

    6.02×10^{22} molecules → x g of $CaCO_3$

    $x = \frac{6.02 \times 10^{22} \times 100}{6.023 \times 10^{23}} = \frac{10^{24}}{10^{23}} = 10$ g

    22.4 g of $H_2 \rightarrow 2g$

    112 L of $H_2 \rightarrow xg$

    $x = \frac{112 \times 2}{22.4} = 10g$

    1 mole of "He" → 4 g

    2.5 moles → x

    $x = 2.5 \times 4 = 10g$

    22.4 L of CO_2 → 44 g

    2.24 L of $CO_2 \rightarrow x$

$x = \frac{2.24 \times 44}{22.4} = 4.4$ g

a. 18 g of water $= 6.023 \times 10^{23}$ molecules

34 g of water $= x$

$x = \frac{34 \times 6.023 \times 10^{23}}{18} = 11.376 \times 10^{23}$

2. 44 g of $CO_2 = 6.023 \times 10^{23}$ molecules

28 g of $CO_2 = x$

$x = \frac{28 \times 6.023 \times 10^{23}}{44} = 3.832 \times 10^{23}$

3. 32 g of $CH_3OH = 6.023 \times 10^{23}$ molecules

46 g of CH_3OH = x

$x = \frac{46 \times 6.023 \times 10^{23}}{32} = 8.65 \times 10^{23}$

4. 108 g of $N_2O_5 = 6.023 \times 10^{23}$ molecules; 54 g of $N_2O_5 = x$

$x = \frac{54 \times 6.023 \times 10^{23}}{108} = 3.0115 \times 10^{23}$

14. Molecular mass in grams of $CaCO_3$; $CaCO_3 = 40 + 12 + 3 \times 16 = 100$ g

$6.022 \times 10^{23} \rightarrow 100$ g; $6.022 \times 10^{22} \rightarrow x$; $x = \frac{6.022 \times 10^{22} \times 100}{6.022 \times 10^{23}}; = 10$ g

Level – III

15) C 16) A 17) D 18) D 19) B 20) C 21) C

22) a-r; b-p; c-q; d-s 23) D 24) A 25) B

15. 6.023×10^{23} Molecules of "He" occupies 22.4 lit

3.03×10^{23} Molecules of "He" occupies x lit

$x = \frac{22.4 \times 3.03 \times 10^{23}}{6.023 \times 10^{23}} = 11.2$lit

16. 64 g of $SO_2 \rightarrow 22.4$lit

3.2 g of $SO_2 \rightarrow$ x lit

$x = \frac{22.4 \times 3.2}{64} = 1.12$ lit

24. 6.023×10^{23} molecules = 18 g of water = ?

1 molecule $= \frac{18}{6.023 \times 10^{23}}$

$= 2.98 \times 10^{-23}$ g

Density = 1 g/cm^3

Volume of 1 molecule of water is

$= \frac{\text{mass of 1 molecule of } H_2O}{\text{Density}}$

$= \frac{2.98 \times 10^{-23}\ \text{g}}{1\text{gcm}^3} = 2.98 \times 10^{-23}\ \text{cm}^3$

25. Mass of electron in amu $= \frac{9.108 \times 10^{-28}\ \text{g}}{1.66 \times 10^{-24}\ \text{g}}$

WORK SHEET-4

Level – I

1) C	2) A	3) D	4) C	5) D	6) B	7) D
8) A	9) D	10) C				

3. 6.023×10^{23} molecules → _44 g

3 molecules → x

$x = \frac{44 \times 3}{6.023 \times 10^{23}}; = 22 \times 10^{-23}$ g

4. 1 Mole of $Ca(OH)_2$ → 74 g

22×10^{-3} Moles of $Ca(OH)_2$ → $xg; x = 74 \times 2.2 \times 10^{-3} = 0.1628$ g

5. $Ca(OH)_2 \rightleftharpoons Ca^{+2} + 2OH^-$

1 Mole of $Ca(OH)_2$ → $3 \times 6.023 \times 10^{23}$ ions

2.2×10^{-3} Moles of $Ca(OH)_2$ → x

$x = 3 \times 2.2 \times 6.023 \times 10^{20}$ ions $; = 39.6 \times 10^{20}$

8. 22.4 L^2 of O_2 is 1Mole

56 L° of O_2 is $x; x = \frac{56}{22.4} = 2.5$

9. 22.4 lit of N_2 is 6.023×10^{23} Molecules; 44.8 lit of N_2 is x

$x = \frac{44.8 \times 6.023 \times 10^{23}}{22.4} = 12.04 \times 10^{23}$

Level – II

1) A	2) B	3) B	4) A	5) A	6) A	7) B
8) D	9) C	10) D	11) C	12) B	13) A	14) A

1. 120 g of $MgSO_4$ → 1 mole

12 g of $MgSO_4$ → x

$x = \frac{12}{120} 0.1$

4. 1 Mole of Ammonia → 17 g

1.5 mole of Ammonia → x

$x = 1.5 \times 17 = 25.5$ g

5. 1 mole of anything has 6.023×10^{23} molecules

2 molecules of Methane (CH_4) contains 8 Hydrogen atoms

2 moles of Methane (CH_4) Contains $8 \times 6.023 \times 10^{23}$ atoms

$= 48.18 \times 10^{23}$

7. The abundance of $C-14$ is 2%

Mass of C - 14 isotope in 12 g sample $= \frac{2\times12}{100} = 0.24$

No. of moles of $C-14 = \frac{0.24}{14} = 0.01714$

No. of atoms $= 0.01714 \times 6.023 \times 10^{23} = 0.103 \times 10^{23}$

8. $3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$

0.3 moles of $BaCl_2$ will react with $\frac{2}{3} \times 0.3; = 0.2$ moles of Na_3PO_4

Now 2 moles of Na_3PO_4 give 1 mole of $Ba_3(PO_4)_2$

0.2 mole of Na_3PO_4 will give $\frac{1}{2} \times 0.2; = 0.1$ mole of $Ba_3(PO_4)_2$

9. At STP $22400CC = 6.023 \times 10^{23}$ atoms

Number of atmos present in one CC

$$\frac{= 6.023 \times 10^{23}}{22400} = 2.69 \times 10^{19}$$

10. Mass one electron $= 9.1 \times 10^{-31}$ kg

Mass of one mole of electrons = mass of N_A electrons $= 9.1 \times 10^{-31} \times 6.023 \times 10^{23}$

5.5×10^{-7} kg

11. 1 Mole of CO contains 6.023×10^{23} atoms of oxygen

6.02×10^{24} "CO" Molecules contain 6.02×10^{24} atoms of oxygen is equal to 10 gram atoms of oxygen

oxygen is diatomic 10 gram atom $= \frac{1}{2} \times 10 = 5$

12. $\frac{6.023\times10^{24}}{6.023\times10^{23}} = 1$ Ogramatoms

13. No. of moles of $CaCO_3 = \frac{50}{100} = 0.5$

1 Mole Contains $\rightarrow 3 \times 6.022 \times 10^{23}$ atoms of oxygen

0.5 Moles $\rightarrow x$

$0.5 \times 3 \times 6.022 \times 10^{23}; = 9.033 \times 10^{23}$

1 Mole of $CaCO_3$ contains 3×16 g of oxygen

0.5 moles $CaCO_3$ contains = xg

$3 \times 16 \times 0.5 = 24g$

14. Molecular weight of $CH_4 = 12 + 4 \times 1 = 16$

No. of moles $= \frac{1.6}{16} = 0.1$

No. of electrons in 1 mole of $CH_4; 6 + 4 = 10$ electrns

1 Mole of CH_4 contains $= 6.023 \times 10^{23} \times 10$ electrons 0.1 mole of CH_4 contains = x $x =$

$6.023 \times 10^{23} \times 10 \times 0.1; = 6.023 \times 10^{23}$

Level – III

15) A, B,C, D 16) A, B, C, D 17) D 18) D 19) C

20) a-s; b-r; c-q; d-p 21) D 22) A 23) A

24) A, B, C, D 25) a-q; b-p; c-r; d-s

21. W = 108 g

GMW OF Glucose = 180 g; $(C_6H_{12}O_6)$

No.of moles $= \frac{108}{180} = 0.6$ moles

22. Density $= \frac{\text{Mass}}{\text{Volume}} = \frac{44}{22.4} = 1.96$ lit $^{-1} = 2$ glit^{-1}

WORK SHEET-5

Level – I

1) C 2) A 3) C 4) A 5) B 6) B 7) A

8) D 9) A

Level – II

1) C 2) B 3) D 4) C 5) B 6) C 7) B

8) A 9) A 10) C 11) C 12) C 13) C 14) B

15) C 16) A

1. $E = \frac{\text{Atomic weight}}{\text{valency}} = \frac{40}{2} = 20$
2. $E = \frac{12}{4} = 3$
3. $E = \frac{23}{1} = 23$
4. $E = \frac{35.5}{1} = 35.5$
5. $E = \frac{27}{3} = 9$
6. $E = \frac{24}{2} = 12$
7. E $= \frac{56}{2} = 28$
8. E $= \frac{63.5}{2} = 31.75$
9. The Mn in $KMnO_4$ exists in +7 state

   In acidic medium Mn^{+7} goes to Mn^{+2} There is a net gian of 5 electrons.

   so E $= \frac{158}{5} = 31.69$
10. The Mn in $KMnO_4$ exists in +7 state

    In basic medium Mn^{+7} goes to Mn^{+6}.

    The difference is of 1 electron.

    so E $= \frac{158}{1} = 158$
11. Equivalent weight of $KMnO_4$ in neutral medium is E $= \frac{158}{3} = 52.66$
12. $E = \frac{16}{2} = 8$
13. $E = \frac{Mwt}{\text{acidity}} = \frac{78}{3} = 26$

14. $E = \frac{Mwt}{\text{acidity}} = \frac{56}{1} = 56$

15. $E = \frac{126}{2} = 63$

16. In Potassium dichromate ($K_2Cr_2O_7$)

Chromium oxidation state is +6 . In acidic medium it changes from +6 to +3 on "Reduction"

$$E = \frac{294}{(2 \times 3e^-)} = \frac{294}{6} = 49$$

Level – III

17) A,B,C 18) C 19) B 20) C 21) a-r; b-q; c-p; d-s

22) a-q; b-p; c-r; d-s 23) C 24) A 25) B

19. $12 = \frac{\text{Atomic weight}}{2} \Rightarrow$ Atomic weight $= 12 \times 2 = 24$

20. $\frac{106.5}{35.5} = 3$ it means ' x ' is combining with ' 3 ' chlorine atoms so valency is 3 .

$$E = \frac{27}{3} = 9$$

23. $29 = \frac{\text{Molecular mass}}{2}$

Molecular mass = 58 g/mol

25. $E = \frac{74}{2} = 37g$

4. SOLUTIONS

WORKSHEET – 1

Level – I

1) A	2) B	3) C	4) A	5) B	6) C	7) D
8) A	9) A	10) B	11) B	12) B	13) B	

Level – II

1) C	2) A	3) B	4) B	5) D	6) D	7) B
8) B	9) A	10) A	11) C	12) D	13) D	14) 1

Level – III

15) A, D	16) B, C, D	17) A	18) C	19) A	20) a- s, b - r, c-p, d-q
21) D	22) A	23) B	24) a - r, b-s, c-p, d-q	25) A, B, C, D	
26) a - r, b-q, c-p					

WORKSHEET – 2

Level – I

1) B	2) D	3) A	4) D	5) A	6) A	7) C
8) D	9) C					

3. $\frac{4}{40} \times 100 = \frac{400}{40} = 10$

4. $M\% = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100 = \frac{10}{12} \times 100 = \frac{50}{6} = 8.33$

Level – II

1) D	2) A	3) A	4) C	5) B	6) D	7) B
8) C	9) C	10) A	11) A	12) C	13) A	14) C
15) C						

3. $\frac{4}{36+4} \times 100; \frac{4}{40} \times 100; = \frac{1}{10} \times 100; = 0.1 \times 100; = 10$

4. $\frac{10}{50} \times 100; = 10 \times 2 = 20$

5. $10 = \frac{x}{(x+300)} \times 100; 10(x + 300) = 100 \times x; 10x + 3000 = 100x$

$10x + 3000 = 100x; 100x - 10x = 3000; \therefore x = \frac{300}{9} = 33.33$

6. $\frac{50}{550} \times 100 = \frac{10}{11} \times 10 = \frac{100}{11} = 9.09$

7. Sugar cane

$25\% = \frac{x}{250} \times 100; 25 \times 2.5 = x; x = 62.5g$

= Total wt. - sugar cane weight; = 250 g – 62.5 g; water wt. = 187.5 g

8. $\frac{10}{200} \times 100; = \frac{10}{2} = 5\%$

9. $\mathrm{d} = \frac{\mathrm{m}}{\mathrm{v}}; \mathrm{v} = \frac{\mathrm{m}}{\mathrm{d}}; = \frac{15}{0.96} \times \frac{100}{100} = \frac{1500}{96} = 15.6$

10. $H_2O\% = 100\% - 3.50\% = 96.5\%$

It means in 100 g of Sea water 96.5 g of H_2O is present

In ' kg ' $= 96.5 \times 10^{-3} = 0.0965$ kg

11. $35 = \frac{\mathrm{wt}}{1.3} \times 10^6$

$\mathrm{wt} = \frac{35 \times 1.3}{10^6}$

$= 45.5 \times 10^{-6}$ g

$= 4.55 \times 10^{-5}$ g.

12. $\mathrm{ppm} = \frac{\text{wt of solute}}{100} \times 10^6; 155 = \frac{\mathrm{x}}{100} \times 10^6; = \frac{155 \times 100}{10^6} = \mathrm{x}; = \mathrm{x} = \frac{15500}{10^6}; \mathrm{x} = 0.0155$ g

13. $\frac{20}{80+20} \times 100; = \frac{20}{100} \times 100 = 20\%$

14. $\frac{15}{150} \times 100 = 5 \times 2 = 10$

15. a) $\frac{10}{190+10} \times 100; = \frac{10}{200} \times 100; = 5$

b) $\frac{30}{200} \times 100 = 15$

Level – III

16) A 17) A 18) C 19) A 20) B 21) a- p, b-q, c-r, d-s

22) 10 23) D 24) A 25) A, B 26) B

16. Salubility

$\mathrm{s} = \frac{\text{wt of solute}}{\text{wt of solvent}} \times 100; = \frac{10}{100} \times 100; \mathrm{s}\% = 10$

$\text{Weight } \% = \frac{\text{wt of solute}}{\text{wt of solute + solvent}} \times 100$

$= \frac{10}{110} \times 100; \frac{100}{11}$

$(\mathrm{w/w})\% = 9.09\%$

17. $= \frac{20}{80+20} \times 100; = \frac{20}{100} \times 100 = 20\%$

18. $\mu_1 = 6\% \; \mu_2 = 30\%$

$v_1 = \mathrm{x} \; v_2 = 200$

$\mu_1 v_1 = \mu_2 v_2; 6 \times x = 30 \times 200; \text{x} = \frac{30 \times 200}{6}; \text{x} = 1000\text{gm}; v_2 - v_1 = 1000 - 200$

$= 800\text{gm}$

19. $20\% = \frac{70}{(70+\text{x})} \times 100; 20(70 + \text{x}) = 70 \times 100; 20\text{x} + 1400 = 7000$

$20\text{x} = 7000 - 1400; 20\text{x} = 5600; \text{x} = \frac{5600}{20}; \text{x} = \frac{560}{2}; \text{x} = 280$

20. $\frac{15}{75} \times 100; = 5 \times 4 = 20$

21. $\text{ppm} = \frac{10}{1000} \times 10^6 = 10{,}000\text{ppm}$

22. $18\% = \frac{\text{x}}{180} \times 100; = \frac{324}{10}; = 32.4$ g(glucose)

Mass of water = Total - glucase weight

$= 180 - 32.4; = 147.6$ g

WORKSHEET – 3

Level – I

1) B	2) B	3) B	4) D	5) D	6) D	7) C
8) C	9) C	10) D	11) A	12) C	13) D	14) C
15) A						

7. $\text{M} = \frac{\text{W}}{\text{GMW}} \times \frac{1000}{\text{V(ml)}}; \text{V} = 100\text{ml}; \text{M} = 0.1\text{M}$

GMW of $Na_2CO_3 = 106; 0.1 = \frac{\text{w}}{106} \times \frac{1000}{100} \Rightarrow \text{w} = 1.06$ g

8. $\text{M} = \frac{\text{n}}{\text{V(lit)}}; \text{V} = 600\text{ml} = 0.61\text{it}; \text{M} = 0.05\text{M}; \therefore \text{n} = \text{M} \times \text{V} = 0.05 \times 0.6 = 0.03$ moles

9. Solution - I

Volume of NaOH solution = $V_1 = 100$ml

Molarity = $M_1 = 0.5$M

Solution - II

Volume of NaOH = $V_2 = ?$

Molarity = $M_2 = 0.2$M

Dilution Law $M_1\ V_1 = M_2\ V_2$

$V_2 = \frac{M_1\ V_1}{M_2} = \frac{0.5 \times 100}{0.2} = 250$ml

Volume of water to be added = $V_2 - V_1 = 250 - 100 = 150$ml

15. $\text{F} = \frac{47.5}{96} \times \frac{1000}{500} = 1$

Level – II

1) B	2) B	3) C	4) A	5) A	6) B	7) C
8) B	9) C	10) D	11) A	12) D	13) C	14) B
15) B	16) B	17) C	18) C	19) B		

1. $M = n \times \frac{1}{V \text{ in lit}} = 0.006 \times \frac{1}{100 \times 10^{-3}} = 0.006 \times \frac{1}{10^{-1}} = 0.06$

2. Wt. of $Na_2SO_4 = w = 7.1$ g

   GMW of $Na_2SO_4 = 142$ g

   Volume of solution = v = 100ml

   $M = \frac{W}{GMW} \times \frac{1000}{Vml} = \frac{7.1}{142} \times \frac{1000}{100}; M = 0.5M$

3. $M = \frac{100 \times 0.1 + 200 \times 0.1}{300} = 0.1M$

4. $5 = \frac{wt}{111} \times \frac{1000}{500}; wt = \frac{555}{2} = 277.5$ g.

5. $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

   $M_1 = 0.1 \quad M_2 = 0.2$
   $V_1 = ? \quad V_2 = 40ml$
   $n_1 = 1 \quad n_2 = 2$

   $\frac{M_1\ V_1}{n_1} = \frac{M_2\ V_2}{n_2}$

   $\frac{0.1 \times V_1}{1} = \frac{0.2 \times 40}{2}; V_1 = 40ml$

7. $(V_2 - V_1) = \left(\frac{M_1 - M_2}{M_2}\right) V_1$

8. Molarity of pure $H_2O = \frac{1000}{18} = 55.55$

9. Dilution Law $M_1\ V_1 = M_2\ V_2$

   $250 \times 0.25 = 500 \times M_2$

   $M_2 = 0.125M$

10. No of moles of urea $n = 0.1$

    Wt of solvent W = 100 g = 0.1Kg

    $\therefore m = \frac{n}{Wkg} = \frac{0.1}{0.1} = 1$ m

11. Wt. of methanol $(CH_3OH) = 16$ g; GMW of $CH_3OH = 32$ g

    Volume of solution (V) = 100ml

    Density of solution (d) = 0.96 g/ml

    Wt. of solution $= v \times d \left(\because d = \frac{m}{V}\right) = 100 \times 0.96 = 96$ g

    $\therefore$ Wt of solvent = W = wt of solution −wt of solute

    $= 96 - 16 = 80$ g

$$m = \frac{w}{GMW} \times \frac{1000}{W(\ g)} = \frac{16}{32} \times \frac{1000}{80} = 6.25\ m$$

12. $M_R = \frac{M_1\ V_1 + M_2\ V_2}{V_1 + V_2}; \frac{250\times6+650\times3}{900} = \frac{1500+1950}{900} = \frac{3450}{900} = 3.83$

$M_R = 3.83,\ V_R = 900ml$; Volume of water $= \left(\frac{3.83-3}{3}\right)900 = 249ml$

13. Mass percentage of sulphuric acid $= x = 13\%$

Density $= d = 1.09$ g/ml

GMW of $H_2SO_4 = 98$

$$M = \frac{10 \times d \times x}{GMW} = \frac{10 \times 1.09 \times 13}{98} = 1.445M$$

14. Wt of Nacl $= W = 50$ g

GFW of Nacl $= 58.5$ g

Wt of solution $= 500$ g

Density of solution $= d = 0.936\ g/cm^3$

$\therefore$ Volume of solution $V = \frac{mass}{Density}$

$\left(\because d = \frac{m}{v}\right)$

$$= \frac{500}{0.936} ml = 534.19ml$$

$$F = \frac{W}{GFW} \times \frac{1000}{V}$$

$$= \frac{50}{58.5} \times \frac{1000}{500} \times 0.936; \left(\therefore V = \frac{500}{0.936} ml\right)$$

$F = 1.6$ F

15. $M = \frac{wt}{GMW} \times \frac{1000}{Vml}; = \frac{0.53}{106} \times \frac{1000}{100}; = \frac{5.3}{106} = 0.05M$

16. $M = \frac{wt}{GMW} \times \frac{1000}{Vml}; 0.1 = \frac{x}{98} \times \frac{1000}{400}$

$$0.1 = \frac{x}{98} \times 2.5; x = 3.92gms$$

17. Find NaCl weight difference in 0.3M and 0.4M Solutions.

18. $d = 1.02$ g/ml; $M = 2.05M$

CH_3COOH mol wt $= 60$

$$m = \frac{1000 \times M}{1000 \times d - M \times Gwt}$$

$$= \frac{1000 \times 2.05}{1000 \times 1.02 - 2.05 \times 60} = 2.28\ mol\ kg^{-1}$$

19. For nutralizations

$N_1\ V_1 = N_2\ V_2; 0.025 \times 2 \times V_1 = 0.03 \times 2 \times 25$

No.of Equivalents (1) = No.of Equivalents (2)

$V_1 = 30ml$

Level – III

20) A,B,C 21) B 22) D 23) C 24) a - p, b-q, c-r, d-s

25) A, B 26) a-p, b-q, c-r, d-p, q, s

21. $M = \frac{d\times \text{ Percent } \times 10}{\text{mol.wt}} = \frac{1.1\times 15\times 10}{98} = \frac{150\times 1.1}{98} = \frac{165}{98} = 1.684M$

22. $M = \frac{1.95\times 10\times 98}{98} = 1.95 \times 10 = 19.5$

23. $M = \frac{98\times 10\times 1.84}{98} = 18.4M$

WORKSHEET – 4

Level – I

1) A 2) A 3) D 4) C 5) D 6) D 7) D

8) A 9) C 10) B

Level – II

1) C 2) C 3) B 4) D 5) B 6) A 7) B

8) B 9) C 10) B 11) A 12) A 13) B 14) D

15) B 16) A 17) B 18) A

1. $M = 0.1M$

   Basicity of H_3PO_3 = n = 2

   $N = n \times M$

   $= 2 \times 1.0 = 2.0$ N

2. Decinormal NaCl = 0.1 NNaCl

   $N = \frac{W}{GEW} \times \frac{1000}{V(ml)}$

   $N = 0.1$ N

   $V = 250ml$

   $GEW = \frac{GMW}{1} = \frac{58.5}{1} = 58.5$

   $0.1 = \frac{W}{58.5} \times \frac{1000}{250}$

   $W = 1.46$ g

3. $\frac{W}{V}\% = Y = 0.98$

   It means 0.98 g of H_2SO_4 present in 100ml of solution

   GEW of $H_2SO_4 = \frac{GMW}{\text{Basicity}} = \frac{98}{2} = 49$

$N = \frac{wt}{GEW} \times \frac{1000}{Vinml} = \frac{0.98}{49} \times \frac{1000}{100}$

$= 0.2$ N.

6. Wt. of NaOH = w = 4.0 g

GEW of NaOH $= \frac{GMW}{Acidity} = \frac{40}{1} = 40$ g

Volume of solution = v = 100ml

$N = \frac{W}{GEW} \times \frac{1000}{vml}$

$= \frac{4}{40} \times \frac{1000}{100}$

N = 1 N

7. Molarity of H_2SO_4 = M = 2M

Normality of H_2SO_4 = N = n × M(Here n = Basicity of H_2SO_4 = 2)

$= 2 \times 2 = 4$ N

8. Amount of oxlalic acid = w = ?

Volume of solution = v = 500ml

Normality of solution = N = 0.1 N

GEW of oxalic acid $= \frac{GMW}{Basicity} = \frac{126}{2} = 63$ g

$N = \frac{W}{GEW} \times \frac{1000}{V(ml)}; 0.1 = \frac{W}{63} \times \frac{1000}{500} \Rightarrow W = 3.15$ g

9. $M_1\ V_1 = M_2\ V_2; 18 \times 5 = M_2 \times 500 \Rightarrow M_2 = 0.18; N = n \times M = 2 \times 0.18 = 0.36$

10. Number of equivalents of HCl = Number of equivalents of NaOH

11. $0.1 = \frac{wt}{100} \times \frac{1000}{100} \Rightarrow wt = 1$ g

12. $N = \frac{W}{GEW} \times \frac{1000}{V(ml)}; 0.1 = \frac{W}{58.5} \times \frac{1000}{250} \Rightarrow W = 1.46$ g

13. Mass percentage of H_2SO_4 = x = 0.98%; Sp.gr. of H_2SO_4 = 1.08

$\therefore N = \frac{10 \times sp.gr \times x}{GEW}; = \frac{10 \times 0.98 \times 1.08}{49} = 0.216$ N

14. $N = \frac{wt}{Eq \cdot Wt} \times \frac{1000}{Vm_1};\ 0.05 = \frac{x}{49.04} \times \frac{1000}{100}; x = \frac{0.05 \times 49.04}{10} = 0.2452$ g

15. $N_1\ V_1 = N_2\ V_2; \frac{20}{40} \times \frac{1000}{x} = \frac{4.9}{49} \times \frac{1000}{400}; x = 2000ml, x = 2$ lit

16. $0.5 = \frac{20}{GMW}; GMW = 40$

17. $N_{H_2SO_4} = M_{H_2SO_4} \times 2; N_{KOH} = M_{KOH} \times 1\ \ N_R = \frac{N_{H_2SO_4} \cdot V_{H_2SO_4} - N_{KOH} \cdot V_{KOH}}{V_{total}}$

$= \frac{0.2 \times 100 - 0.1 \times 100}{2 \times 1000} = \frac{10}{2 \times 10^3} = 5 \times 10^{-3}$

18. $N \times V_{m_1}(\text{ Base }) = \frac{W}{GEW} \times 1000; 0.125 \times 20 = \frac{0.25}{GEW} \times 1000; GEW = 100$

Level – III

19) B, C, D 20) B 21) A 22) B 23) a-p, b-q, c-r, d-t

24) 4 N 25) a - r, b- p, c- s, d- q 26) a -t, b-p, c-q, d-r

WORKSHEET – 5

Level – I

1) C 2) D 3) C 4) C 5) B 6) A 7) C

8) D

8. $\chi_{C_2H_5OH} = \frac{n_{C_2H_5OH}}{n_{C_2H_5OH}+n_{H_2O}}$;

$0.6 = \frac{\frac{69}{46}}{\frac{69}{46}+n_{H_2O}}; 0.6 = \frac{1.5}{1.5+n_{H_2O}}$

$n_{H_2O} = 1$; Wt. of H_2O = 18gm.

Level – II

1) 3 2) 2 3) 2 4) 3 5) 1 6) 1 7) 2

8) 3 9) 1 10) 3 11) 1 12) 1 13) 3 14) 1

15) 2 16) 2

1. $20\% \longrightarrow \frac{20}{100}$; Wt.of H_2O = 80gms; Wt. of H_2O_2 = 20; $X_{H_2O} = \frac{n_{H_2O}}{n_{H_2O}+n_{H_2O_2}}$
2. No. of moles of solute = n_1 = 6 moles; No. of moles of solution = $n_1 + n_2$ = 100 moles

   Mole fraction of solute = $X_1 = \frac{n_1}{n_1+n_2} = 0.06$
3. Wt. of urea = W_1 = 60 g; GMW of urea = M_1 = 60 g; Wt of water = W_2 = 90 g

   GMW of water = M_2 = 18 g; No. of moles of urea = $n_1 = \frac{W_1}{M_1} = \frac{60}{60}; n_1 = 1$

   No. of moles of water = $n_2 = \frac{W_2}{M_2} = 5$

   ∴ mole fraction of solute (urea) $X_1 = \frac{n_1}{n_1+n_2} = \frac{1}{1+5} = \frac{1}{6}$
4. Mass of solvent given = 1 kg = 1000 g; Molecular mass of ethanol=46

   Molecular mass of water = 18; Mole fraction of ethanol = 0.20

   Mole fraction of ethanol = $\frac{n_1}{n_1+n_2}; 0.20 = \frac{\frac{wt}{46}}{\frac{wt}{46}+\frac{1000}{18}} \Rightarrow$ wt = 638.88

5. Mole fraction of $CH_3OH = X_A = 0.02$; Density = d = 0.994

Molality = m = $\frac{X_A \times 1000}{(1-X_A)GMW_B}$; = $\frac{0.02 \times 1000}{(1-0.02) \times 18}$ = 1.13 m

6. Molality = m = 2.5 m

Mole fraction of solute = X_A = ?

GMW of water = GMW_B = 18 g; m = $\frac{X_A \times 1000}{(1-X_A)GMW_B}$; $2.5 = \frac{X_A}{1-X_A} \times \frac{1000}{18}$

$X_A = 0.043$

7. Mole fraction of $CH_3OH = X_A = 0.02$; Density = d = 0.994 g cm^{-3}

GMW of $CH_3OH = GMW_A = 32$ g; GMW of water = GMW_B = 18 g

Molarity = M = ?

$$M = \frac{X_A \times 1000 \times d}{X_A(GMW_A) + X_B(GMW_B)}; = \frac{0.02 \times 1000 \times 0.994}{(0.02 \times 32) + (0.98 \times 18)} = 1.0875M$$

8. Volume of H_2SO_4 = V = 100ml = 0.1 lit; Molarity M = 0.1M. No. of moles of H_2SO_4 = n = ?

$M = \frac{n}{v(lit)}$; $n = M \times V(\text{ lit }); n = M \times V(lit); 0.1 \times 0.1 = 0.01$ mole

Required equation

$Na_2CO_3 + H_2SO_4 \rightarrow Na_2SO_4 + H_2O + CO_2 \uparrow$; 1 mole H_2SO_4 produce 22.4 lit of CO_2

0.01 mole$_2SO_4$ produce 0.224 lit.

9. Mole fraction of water = $X_B = 0.85$; Mole fraction of $H_2SO_4 = X_A = 1 - 0.85 = 0.15$

($\therefore X_A + X_B = 1$); Molality m = ?; m = $\frac{X_A \times 1000}{X_B \times GMW_B}$

= $\frac{0.15 \times 1000}{0.85 \times 18}$ ($\therefore$ GMW of water $GMW_B = 18$); m = 9.80 m

10. Wt of CH_3OH = 48 g

$n_{CH_3OH} = \frac{48}{32} = \frac{3}{2}; X_{CH_3OH} = 0.6; X_{H_2O} = 1 - 0.6 = 0.4$

$X_{H_2O} = 0.4 = \frac{n_{H_2O}}{n_{H_2O} + n_{CH_3OH}}; = n_{H_2O} = 1$; Wt . of H_2O = 18gms

11. Mass of H_2O = 36; Mass of CH_3CHO = 64

$$n_{H_2O} = \frac{36}{18} = 2; n_{CH_3CHO} = \frac{64}{44} = \frac{32}{22}; X_{CH_3CHO} = \frac{\frac{32}{22}}{2 + \frac{32}{22}} = \frac{1.45}{2 + 0.45} = 0.42$$

12. $n_A = \frac{w_A}{M_A} = \frac{138}{46} = 3; n_B = \frac{w_B}{M_B} = \frac{72}{18} = 4; X_A = \frac{n_A}{n_A + n_B} = \frac{3}{3+4} = \frac{3}{7}$

$X_B = \frac{n_B}{n_A + n_B} = \frac{4}{3+4} = \frac{4}{7}; \frac{X_A}{X_B} = \frac{3/7}{4/7} = \frac{3}{4}$

15. w/w% of NaOH = 10%; wt. of NaOH = 10 g; wt. of H_2O = 90 g; $n_{NaOH} = \frac{10}{40} = 0.25$

$$n_{H_2O} = \frac{90}{18} = 50; X_{NaOH} = \frac{n_{NaOH}}{n_{NaOH} + n_{H_2O}} = \frac{0.25}{0.25 + 5} = 0.0476$$

16. $X_A = 0.3; X_B = 0.7 [X_A + X_B = 1]$; % of $X_B = 0.7 \times 100 = 70$

Level – III

17) B, D 18) a - p, b-q, c-r, d-p, s 19) D 20) A 21) B

22) a - q, b-p, c-t, d- r 23) C, D

WORKSHEET – 6

Level – I

1) C 2) A 3) A 4) B 5) B 6) B 7) A

8) A 9) C 10) D

Level – II

1) C 2) D 3) A 4) B 5) B 6) D 7) A

8) A 9) D 10) C 11) D 12) A 13) C 14) C

15) A 16) D 17) D 18) A 19) B

2. $S = 40; W_1 = ?; (W - W_1) = 50; 40 = \frac{W_1}{50} \times 100; W_1 = 20gm$

3. $S = 50,\ W_1 = 90gm.; 50 = \frac{90}{\text{Mass of Solvent}} = \times 100 = 180gm.$

4. $S = X_1\ \ W_1 = Ygm; X = \frac{Y}{W-Y} \times 100; X(W - Y) = 100Y; XW - XY = 100Y;$

   $XW = 100Y + XY; W = \frac{(100 + X)Y}{X}$

8. $S = \frac{\text{Mass of solute}}{\text{Mass of Solvent}} = \frac{2.5}{25} \times 100 = 10.$

9. Wt. of solute = 50 g. ; Wt. of solvent = 100 − 50 = 50 g.; $S = \frac{50}{50} \times 100 = 100$

10. Wt. of solute = 5 g. ; Wt. of solvent = 30 − 5 = 25 g. ; $S = \frac{5}{25} \times 100 = 20$

11. at 30°C

    at 80°C

    $S = 30$

    $S = 50$

    Mass of solvent = 50

    Mass of solvent = 50

    Mass of solute = ?

    Mass of solute = ?

    $30 = \frac{W_1}{50} \times 100$

$50 = \frac{W_1}{50} \times 100$

$W_1 = 15$gm.

$W_1 = 25$gm.

Weight of solute comes out $= 25 - 15 = 10$gm.

18. $P = K_H.C; P = 2.4$ atm; $K_H = 29.76$ atm/mole/L.; $C = \frac{P}{K_H} = \frac{2.4}{29.76} = 0.08$ mol/L.

Weight of $CO_2 = 0.08 \times 44 = 3.52$gm.

Level – III

20) A,B,D 21) A,C,D 22) D 23) C 24) D 25) D

26) a - p, s; b-q,r; c-q, r; d-p,s 27) A,B,D 28) A, B